



Review Article

Nanomaterial-based sensors for heavy metal ions analysis

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ABSTRACT

Heavy metal ions (HMIs) are one of the main causes of environmental pollution, which has become a global epidemic. The rapid, sensitive and reliable HMIs analysis is urgently needed due to their serious threats to human health and the environment. Due to their unique physical and chemical properties, nanomaterials provide more opportunities for HMIs analysis. To date, many efforts have been made to develop nanomaterial-based sensors for HMIs detection. These nanosensors offer several advantages including high sensitivity, selectivity, portability and on-site detection ability. Here, various nanomaterial-based sensors including optical, electrochemical and electronic sensors for HMIs detection are outlined. The roles of nanomaterials in the optical, electrochemical and electronic sensors are highlighted, and the examples and comparisons of these nanosensors for HMIs detection are provided. Furthermore, to meet the on-site detection, portable sensors are discussed.

1. Introduction

With the rapid development of modern agriculture and industry, the environmental problems caused by HMIs are becoming a global epidemic. The main sources of heavy metals released into environment are due to natural means and human activities, including rock and soil erosion and weathering, mining, industrial effluents, pest control agents for crops, and so on [1] [2] [3]. As, Hg, Pb, Cu, Cd, Ni and Cr are classified among the most toxic heavy metals and can cause numerous health issues or toxic effects to human health even at trace levels due to their prolonged half-life, nonbiodegradability, and accumulated distribution in different parts of the body [4] [5] [6]. Some heavy metals play important biochemical functions with trace amounts, but are toxic to organisms and have threat to human health at higher concentrations [7] [8]. Therefore, it is highly desired to monitor HMIs.

To date, many well-matured instrumental analysis technologies have been developed for the detection of HMIs, including atomic absorption spectroscopy, inductively coupled plasma-mass spectrometry, inductively coupled plasma-atomic emission spectrometry, ion chromatography ultraviolet-visible spectroscopy, X-ray fluorescence spectroscopy, etc. [9] [10] [11] [12]. These techniques are accurate and sensitive for HMIs detection and can measure multiple ions simultaneously.

However, their complex operation procedures, expensive costs, professional labor, complicated equipments, and long response time limited their practical applications for portable on-site detection [13] [14] [15]. Therefore, facile, inexpensive, time saving and high throughput procedures are necessary for HMIs detection.

Fortunately, optical, electrochemical, and electronic techniques have been developed as powerful sensing tools for HMIs detection [14] [16] [17] [18] [19] [20] [21]. Optical sensors are based on changes in the optical signals of the optical sensing probes. Electrochemical sensors convert information about the concentration of a target molecule into the intensity of an electrical signal. Electronic sensors, typically field-effect transistor (FET) sensors, are based on the electrical characteristics of transistors and the conductive properties of target analytes, and when target analytes approach the detector, electric field distribution around them are changed, resulting in the altered conductivity of transducer and therefore electrical signals. These techniques offer several advantages such as accurateness, simplicity, low cost, remarkable sensitivity, effectiveness in the multiplexed detection and on-site detection ability.

With the rapid development of nanoscience and nanotechnology, various nanomaterials with unique structures and excellent optical/catalytic/electrical properties have been prepared and widely used in

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optical/electrochemical/electronic sensors for HMIs detection. The nanomaterials involved in sensors have greatly enhanced their reproducibility and detection limit for selective determination of HMIs. For optical sensors, nanomaterials are commonly used as sensing probes due to their optical properties or as catalysts that trigger reactions and therefore result in optical signal changes. While for electrochemical sensors, nanomaterials usually play the roles as electrode modifiers or micro/nanoelectrodes owing to their catalytic and electrical properties. In addition, the electrical properties of semiconductor nanomaterials making them popular choices as transducers for FET sensors.

Several reviews on nanomaterials for heavy metals detection have been reported. For instance, Li and coworkers provided a review on nanostructured sensors for heavy metals detection, in which they highlighted the strategies for design of nanostructured sensors and the benefits of the use of nanomaterials and nanostructures [22]. Liu et al. presented a comprehensive survey of the applications of nanomaterial-based sensors for heavy metals analysis, mainly mentioning a series of nanomaterials synthesized through various chemical and green sources [23]. Azzouz and colleagues systematically summarized nanomaterials-aptasensors along with discussions on their sensing mechanisms and sensing performances for the detection of heavy metal/metalloid ions [24]. Here, we focused our attention on the roles of nanomaterials in the optical, electrochemical and electronic sensors, and the applications of these nanosensors for HMIs detection (Scheme 1).

2. Nanomaterial-based optical sensors

With the development of nanoscience and nanotechnology, various nanomaterials such as gold nanoparticles (AuNPs), gold nanorods (AuNRs), silver nanoparticles (AgNPs), quantum dots (QDs), graphene (G) and graphene oxide (GO) with different optical properties and catalytic activities have been successfully synthesized. On account of their optical properties and catalytic activities, nanomaterials can be used as sensing probes or catalysts for the construction of optical sensors. Apparently, optical nanosensors are based on the optical signal changes of the sensing system caused by the changes of aggregation states or morphologies of nanomaterials or the reactions of the sensing system triggered by catalytic nanomaterials. And according to different optical properties, the optical nanosensors can be classified into colorimetric, fluorescence, surface enhancement of Raman scattering (SERS), and electrochemiluminescence (ECL) nanosensors.

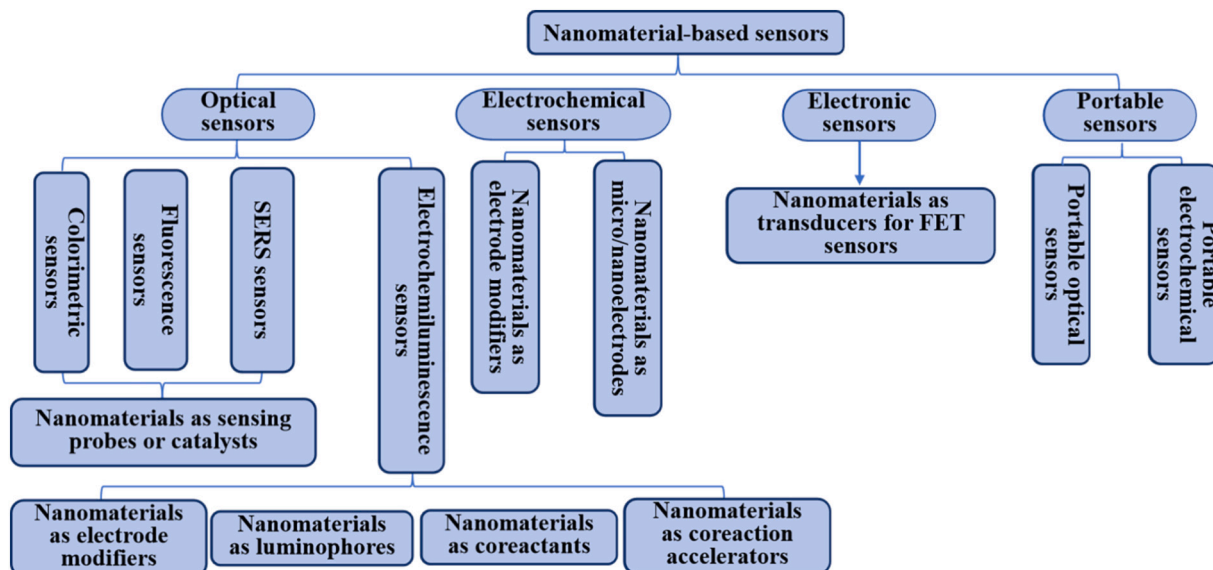
2.1. Colorimetric nanosensor

Colorimetric assay is based on the naked-eye sensitive color change of the detection system before and after colorimetric reaction. It is extremely attractive due to its advantages of rapidity, cost-effectiveness, and simplicity [25]. Nanosensors used for colorimetric analysis of HMIs can be designed based on the localized surface plasmon resonance (LSPR) properties of noble metal nanomaterials or the enzyme-mimicking properties of nanozymes.

2.1.1. Noble metal nanomaterials as sensing probes

Due to their unique LSPR, noble metal nanomaterials, especially gold and silver nanomaterials, show higher molar extinction coefficients in the visible region, which facilitate their widespread use as optical-translating units [26]. In addition, the LSPR frequency greatly depends on the dispersion state or morphology of nanomaterial, and any change in the dispersion state or morphology (size or shape) can affect the LSPR frequency, along with the color change of the nanomaterial dispersion [27]. For example, the color of AgNPs is generally yellow in their monodispersed state, which changes to brown in their aggregated state [28]; the color of AuNPs in aqueous solution changes from brown, orange, red to purple as the size of AuNPs increases from 1 to 100 nm [29]; the color of AuNRs depends on the change of length-diameter ratio [30]. As the LSPR signal changes along with the aggregation state or morphology change of nanomaterials, the colorimetric nanosensor can be divided into two types: one is aggregation state-based method, the other is morphology transition-based method.

For aggregation state-based colorimetric method, target analytes can directly or indirectly induce the interparticle distance-mediated color change of noble metal nanomaterials. Nanomaterials used for aggregation state-based colorimetric method can be labeled or label-free, and target analytes can be detected based on the aggregation or anti-aggregation of labeled or label-free nanomaterials. For labeled nanomaterials, sulfhydryl group is the primary one for surface modification of nanomaterials to make them stabilized and monodispersed, which can be attributed to the strong Au—S and Ag—S bond. Once the introduction of target analytes, nanomaterials aggregate due to the formation of interparticle complexes between the labeled ligands and target analytes, resulting in a color change of the detection system. For instance, Sung and Wu modified AuNPs with N-1-(2-mercaptoethyl)adenine by Au—S bond, and the obtained AuNPs could aggregate in the presence of Ag⁺ by forming Adenine-Ag⁺-Adenine complexes through a bond between the



Scheme 1. Overview of the roles of nanomaterials in the nanomaterial-based sensors.

nitrogen atoms on adenine with Ag^+ , causing an obvious change of color and UV-Vis absorption spectra. The developed method showed specific selectivity for Ag^+ compared with other metal ions and high sensitivity for Ag^+ with a concentration of 3.3 nM [31]. For label-free nanomaterials, avoiding surface modification procedure compared with labeled nanomaterials, it may be more attractive because of their time-effective property. For aggregation, target analytes would directly interact with the label-free nanomaterials. In a recent study, Bhattacharyya and Hossain designed a label free colorimetric assay for selective and sensitive detection of Cd^{2+} based on the aggregation of *Heliotropium indicum* leaf extract mediated AuNPs [32]. FTIR spectral analysis was performed to explain the mechanism of colorimetric response, the results indicated that hydroxyl and carbonyl groups on the surface of AuNPs coordinated with Cd^{2+} , leading to the aggregation of AuNPs. For anti-aggregation, the aggregation of label-free nanomaterials would be triggered by additional introduction of chemical reagents that interact with label-free nanomaterials, which would be prohibited by the introduction of target analytes. For instance, Sun and coworkers developed a facile colorimetric method for Hg^{2+} based on the anti-aggregation of AuNPs by Hg deposition [33]. As depicted in Fig. 1A, the negatively charged citrate-AuNPs were monodispersed and red in color, which aggregated after the introduction of positively charged CTAB due to the strong electrostatic attraction, exhibiting a blue color. When citrate-AuNPs was mixed with Hg^{2+} that was then reduced to Hg by citrate ion, the generated Hg deposited onto the surface of AuNPs, inhibiting the aggregation of AuNPs due to breaking the electrostatic attraction between citrate-AuNPs and CTAB. Based on this sensing strategy, the lowest detectable concentration of Hg^{2+} was 11.9 nM on the basis of the measurement of A650/A520.

For morphology transition-based colorimetric method, the morphologies of noble metal nanomaterials can be changed by particular analyte-induced chemical reactions on their surfaces and the related nanomaterials involves AuNPs, AgNPs and AuNRs. For these

nanomaterials, their size can be enlarged or minished or their shape can be changed by the growth or etching of them. For instance, Pallares and coworkers designed a novel colorimetric sensor for lanthanides based on the anti-growth of AuNPs [34]. In this study, researchers found that the octadentate hydroxypyridinone chelator 3,4,3-LI(1,2-HOPO) can trigger Au^{3+} complexation which can be then reduced to Au^0 and subsequent forming AuNPs. The competition between Au^{3+} and lanthanides that with higher binding affinities toward 3,4,3-LI(1,2-HOPO) inhibited the growth of AuNPs. By this sensing principle, lanthanides were detected with a limit of detection (LOD) of 3 μM which is necessary to quantify lanthanides in waste from industrial processes. Xu et al. proposed a simple and sensitive method for the detection of Cu^{2+} by anti-etching of AuNRs [35]. Horseradish peroxidase (HRP) catalyzed the reaction of H_2O_2 and 3,3',5,5'-Tetramethylbenzidine (TMB) to produce oxTMB which etched AuNRs. However, the formation of Cu^+ from the reaction between Cu^{2+} and sodium ascorbate caused the deactivation of HRP and subsequently the decrease of oxTMB, therefore, the etching of AuNRs was prohibited, accompanying a color change from yellow to light orange. With the increase of Cu^{2+} concentration from 0 to 1.6 nM, a good linear relationship was obtained and the LOD was 0.28 nM. Moreover, the satisfying recoveries were realized over 92 % in river water samples by using the proposed method.

2.1.2. Nanozymes as catalysts

Nanozymes, one sort of emerging nanomaterials with enzyme-mimicking properties, have attracted a great deal of attention toward colorimetric assays due to their ability to catalyze reactions that cause color changes. With the existence of HMIs, the catalytic activities of nanozymes can be enhanced or inhibited, leading to the color change of the detection system.

For instance, Xia et al. reported a novel colorimetric assay for quantifying Pb(II) based on the inhibitory impact of Pb(II) on the oxidase-like activity of octahedral silver oxide (Ag_2O) [36]. In the

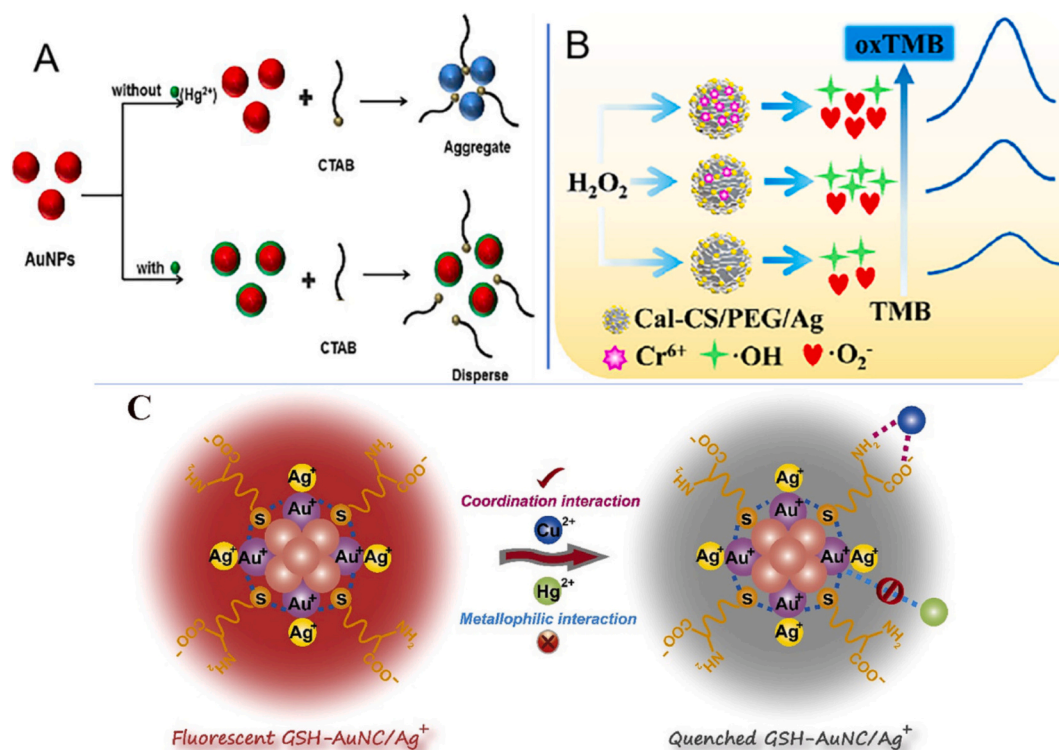


Fig. 1. Schematic illustration of (A) the detection of Hg^{2+} based on the anti-aggregation of AuNPs as the deposition of Hg onto the surface of AuNPs (Reprinted with permission from [33], Copyright (2018) Elsevier), (B) the detection of Cr^{6+} based on the Cr^{6+} -increased peroxidase-like activity of AgNPs (Reprinted with permission from [37], Copyright (2022) American Chemical Society) and (C) the detection of Cu^{2+} and Hg^{2+} based on the quenching of GSH-AuNCs/Ag⁺ (Reprinted with permission from [38], Copyright (2019) Elsevier).

reaction system, the $\bullet\text{O}^{2-}$ produced by the octahedral Ag_2O can catalyze the mixed substrate of 4-aminoantipyrine (4-AAP) and N-ethyl-N-(3-sulfopropyl)-3-methylaniline sodium salt (TOPS) to produce dark purple product. The researcher found that Pb(II) can adsorb onto the surface of octahedral Ag_2O , which hindered the production of $\bullet\text{O}^{2-}$ in the reaction system, resulting in the suppression of oxidase-like activity, and ultimately leading to a lighter purple appearance of the colorimetric reaction solution. Based on this sensing principle and analyzed by a smartphone, Pb(II) can be detected sensitively and selectively. In addition, a visual sensor for Cr^{6+} was successfully constructed by Liu and coworkers based on the Cr^{6+} -increased peroxidase-like activity of AgNPs that were anchored efficiently on carbon spheres (Cal-CS/PEG/AgNPs) [37], as shown in Fig. 1B. The enhanced peroxidase-like activity can be explained by the calculation results of density functional theory which indicated that the addition of Cr^{6+} not only adjusted the electronic structure of Ag atoms but also improved the geometric structure of the adsorbed intermediate, resulting in the optimization of free energy and change of bond lengths in the catalytic reaction process. By using the proposed method, a linear relationship was obtained in the Cr^{6+} concentration range of 0.09–20 μM and the LOD was calculated to be 79 nM.

2.2. Fluorescence nanosensor

Fluorescence sensor is an emerging method from recent decades and has aroused wide concerns due to its high sensitivity and selectivity with quick response times. And it can translate the target recognition information into changeable fluorescence signals to detect target analytes. Typically, a fluorescence nanosensor is composed of a fluorescent component that signals the binding event and a receptor element for specific binding to analytes. Organic dyes are the commonly used fluorescent materials that suffer from the shortcomings of complicated synthesis and modification procedures and poor water solubility. With the development of nanotechnology, nanomaterials are promising alternatives due to their unique optical and physicochemical properties, offering distinct superiority over organic dyes for target analyte sensing [39]. Metal nanoclusters (MNCs) and quantum dots (QDs) are the primary probes for fluorescent sensing [40] [41] [42] [43] [44] [45].

MNCs are water-soluble molecular aggregates composed of dozens of atoms that are transitional forms between atoms and large nanoparticles, and they can be synthesized by etching of nanoparticles (top-down) or assembling of atoms (bottom-up). The application of MNCs in sensing of HMIs has aroused wide concerns due to their ease of fabrication and surface modification, high biocompatibility, and especially fascinating photoluminescence properties. As MNCs are ultrasmall, they could easily aggregate into large nanoparticles without the protection of proper ligands, therefore, surface ligands are necessary for their stabilization and dispersion. Peptides, proteins, oligonucleotides and dendrimers are the commonly used ligands in the synthesis of MNCs [46]. MNCs-based fluorescent nanosensors have been reported for the detection of various HMIs.

QDs are nanocrystals with a size typically between 1 and 10 nm that offer more scope to be utilized by the HMIs sensing due to their fluorescence properties. The fluorescence properties of QDs can be adjusted by changing the composition, morphology, size, doping, and surface engineering. Therefore, QDs have been applied successively to fluorescence sensors based on their excellent properties, such as good photostability, easy surface modification and optical adjustment. Similar to MNCs, surface ligands are also necessary for QDs to ensure their stabilization. And QDs can be surface functionalized easily by conjugation electrostatically or covalently, either directly or via a bridge [47]. QDs-based fluorescent nanosensors have also been reported for the detection of various HMIs.

The recognition of HMIs by using MNCs or QDs as fluorescent probes are primarily based on the quenching or enhancement of fluorescent signals which is resulted from the strong metallophilic interaction between HMIs and the metal nuclei of MNCs, the interaction between HMIs

and ligands of MNCs or QDs coupled with electron/energy transfer effect, or the interaction between HMIs and quenching agents.

2.2.1. Interaction between HMIs and metal nuclei of MNCs

A great deal of studies have been reported for Hg^{2+} detection by using MNCs as fluorescent probes. Based on the strong metallophilic bonds between $5d^{10}$ centers of Hg^{2+} and Au^+ ions, AuNCs stabilized with various ligands have been prepared for Hg^{2+} detection. Sahu and coworkers synthesized L-cysteine assisted chitosan reduced gold nanoclusters (L-cys/CS-AuNCs) through a facile one-pot synthetic protocol, achieving quick response time of 6 min and high sensitivity with LOD of 3 nM for Hg^{2+} detection. The practical applicability of this sensor was verified by application in different water samples with good recoveries of 95.7 %–100.1 % [48]. Similar to this principle, chicken egg white-stabilized AuNCs [49], bovine serum albumin and bromelain protected AuNCs [50], N-Cholyl-L-Cysteine encapsulated AuNCs [51], DNA-templated AuNCs [52], Ce-aggregated 6-aza-2-thiothymine stabilized AuNCs [53] etc. have been synthesized to detect Hg^{2+} .

2.2.2. Interaction between HMIs and ligands

In addition, the interaction between HMIs and ligands of MNCs or QDs have been widely used for various HMIs sensing. Glutathione (GSH) is a widely used stabilizer as well as a reducing agent due to its special coordination interaction with HMIs. Bian et al. reported the GSH-stabilized AuNCs (GSH-AuNCs) as fluorescent probe to detect Hg^{2+} and Pb^{2+} on account of the aggregation-induced fluorescence quenching or enhancement properties of AuNCs, respectively [54]. In this work, Hg^{2+} was detected based on the metallophilic Hg^{2+} - Au^+ interactions and Pb^{2+} was detected based on the high affinity of electrostatic and coordination interactions between Pb^{2+} and the monovalent carboxylic anions from GSH. The GSH-AuNCs based test paper showed the lowest detectable concentration for Hg^{2+} of 5 μM and Pb^{2+} of 50 μM , respectively. Similar to this, GSH-AgNCs have been reported for the detection of Pb^{2+} with a LOD of 0.2 nM, which can meet the requirements of Pb^{2+} analysis in drinking water [55]. In addition, a fast and effective method for sensitive and selective detection of Cu^{2+} have been reported by using GSH-AuNCs based on the interaction between Cu^{2+} and GSH [38]. In the presence of Hg^{2+} , the detection of Cu^{2+} will be interfered. Promisingly, with the help of Ag^+ , Cu^{2+} can be sensitively detected, which can be explained by the fact that Ag^+ can not only enhance the fluorescent signal but also mask the interference of Hg^{2+} due to the stronger Ag^+ - Au^+ metallophilic bond compared with the Hg^{2+} - Au^+ metallophilic bond, as depicted in Fig. 1C. Besides GSH, other ligands such as mercaptosuccinic acid (MSA) [56], mercaptopropionic acid [57], 1,4-diaminobutane (DAB) [58] showing high coordination interaction with HMIs are also used to stabilize MNCs or QDs. For instance, MSA is recognized as a stabilizer of choice that can provide QDs with high stability and good luminescence performance as well as having two effective -COOH functional groups giving strong affinity to multiple species. Ajroud and coworkers prepared MSA capped CdTe QDs to detect Hg^{2+} cations based on the quenching of fluorescent signals which can be ascribed to the electrostatic interaction between Hg^{2+} and MSA followed by electron transfer from QDs excited states to mercury ions and competitive cations exchange [56]. Based on the same mechanisms, Sharma and coworkers synthesized crystalline DAB-capped MoS_2 QDs by a facile hydrothermal method to detect Pb^{2+} [58]. Results indicated that the complex between amine-functionalized MoS_2 QDs and Pb^{2+} were formed and the electron transfer from the conduction band of QDs to the electron-deficient Pb^{2+} ions was the major factor of the photoluminescence quenching of the MoS_2 QDs. Shi et al. designed a novel oxygen-rich N-doped GQDs (N-OGQDs) as fluorescent probe to detect Hg^{2+} based on the fluorescence quenching which can be attributed to the electron-transfer mechanism [59]. N-OGQDs possess a short fluorescent lifetime that can perform fluorescence resonance energy transfer between the N-OGQDs and quencher, in which N-OGQDs act as a donor to sensitize HMIs. The stronger affinity of Hg^{2+} with O atoms of the N-OGQDs brings them in

close proximity with each other, and nonradiative electron transfer occurs from the excited states of N-OGQDs to the d orbital of Hg^{2+} , leading to fluorescence quenching.

2.2.3. Interaction between HMIs and quenching agents

HMIs can induce the recovery of the fluorescence that quenched by quenching agents due to the interaction between them. The initial bright fluorescence of CdTe QDs can be effectively quenched by sulfur anions (S^{2-}), and the presence of Zn^{2+} or Cd^{2+} can “turn-on” the weak fluorescence of CdTe QDs quenched by S^{2-} due to the formation of ZnS or CdS passivation shell. Based on this mechanism, a CdTe QDs-based sensitive and selective probe for Zn^{2+} and Cd^{2+} in aqueous media was reported by Xu and coworkers [60]. The proposed method showed good linear relationship between the fluorescence response and concentration of Zn^{2+} (1.6–35 μM) and Cd^{2+} (1.3–25 μM), and LOD of 1.2 and 0.5 μM for Zn^{2+} and Cd^{2+} , respectively. Radhakrishnan et al. reported the carbon dots@graphitic-carbon nitride ($\text{CDs@g-C}_3\text{N}_4$) nanocomposite as fluorescent probe for the detection of multiple HMIs [61]. The fluorescence intensity of CDs was quenched via nonradiative energy transfer from CDs to g- C_3N_4 nanosheets. Then, the additional HMIs would form a strong surface complex with g- C_3N_4 nanosheets and change the surface properties of g- C_3N_4 nanosheets to release the CDs to “turn-on” fluorescence signal.

2.3. SERS nanosensor

SERS is a high-sensitivity spectroscopic analysis technique that adsorbs target analytes on the surfaces of rough metal to enhance their Raman signals on the basis of conventional Raman spectroscopy analysis technique. And it has been regarded as a promising new technique for the determination of trace HMIs on account of its special fingerprint identification capability, high sensitivity, in-situ and non-destructive detection ability, and simple operation. The elaborated theory of SERS is attributed to the electromagnetic enhancement related to the LSPR on the adjacent noble metal nanostructures and the chemical enhancement caused by charge transfer between the SERS substrates and adsorbed molecules [62] [63] [64]. Various noble metal nanomaterials have been reported as SERS substrates due to their excellent enhancement of Raman signals. The Raman signals can be directly detected through the stretching vibration of covalent bonds of heavy metal oxoanions or indirectly detected through the changes in spectral characteristics during the formation of ligand-metal ion complexes.

2.3.1. Direct detection

Some HMIs exist as their oxoanions in water, such as uranium (VI) exists as uranyl ion, arsenic (III) exists as arsenite (AsO_3^{3-}), arsenic (V) exists as arsenate (AsO_4^{3-}), chromium (VI) exists as chromate (CrO_4^{2-}), these spatially structured compounds possess Raman activity. Therefore, their enhanced Raman signals can be directly detected through the stretching vibration of their covalent bonds by using noble metal nanomaterials as SERS substrates.

Wang and coworkers prepared uniform and compact AgNPs as SERS substrate by self-assembly strategy. Uranyl ions could be adsorbed on the as prepared substrate, which caused the blue shift of stretching vibration frequency of $\text{O}=\text{U}=\text{O}$ from 871 cm^{-1} to 720 cm^{-1} and 826 cm^{-1} derived from oxy-bridge and surface coordination between uranyl and silver substrate, respectively. This sensing strategy showed a linear relationship to uranyl ions concentration in the range of 10^{-7} – 10^{-3} M with a LOD of 100 nM and good reproducibility ($\text{RSD} = 3.27$, $n = 8$) [65]. In order to improve the adsorption ability of uranyl ions on substrates and thus further enhance its Raman intensity, organic ligands that strongly coordinated with uranyl ions have been used to modify the surface of noble metal nanomaterials. (Aminomethyl)phosphonic acid (APA) was utilized to modify AuNPs [66], the stretching vibration frequency of $\text{O}=\text{U}=\text{O}$ shifted to 830 cm^{-1} due to the coordination reaction between uranyl and phosphonic acid, and the Raman intensity was

proportional to the concentration of uranyl ions. The SERS spectra of uranyl ions exhibited excellent reproducibility and the LOD of 800 nM for uranyl ions was achieved.

Arsenic and chromium contaminations in drinking water have attracted considerable attentions. As shown in Fig. 2A, Lv et al. found that surface silver ions can drive arsenite (As(III)) and arsenate (As(V)) adsorbed on the surface of AgNPs to generate amplified SERS signals due to the low solubility product of silver arsenite/silver arsenate [67]. The SERS signals of As(III) at 457 cm^{-1} , 725 cm^{-1} could be observed using AgNPs@Ag^+ as substrate, while As(V) at 798 cm^{-1} could be observed. The linear relationship for As(III) and As(V) were determined as in the range of 9.4×10^{-8} – 3.8×10^{-6} M and 1.4×10^{-7} – 7.0×10^{-6} M, respectively. And LOD for As(III) and As(V) were determined as 4.0×10^{-11} M and 8.0×10^{-11} M, respectively. Zhou and coworkers prepared silver nanowire (AgNWs) arrays as SERS substrate for the detection of Cr (VI) [68]. They found that Cr(VI) showed characteristic peak at 788 cm^{-1} , which is attributed to the stretching vibration of Cr–O bond. And with the decrease of Cr(VI) concentrations, the characteristic peaks gradually red shift, which may be due to the different orientation of the adsorbed molecules as Cr(VI) is parallel to and tilting away from the AgNWs surface at low and high concentrations. The as designed SERS sensor showed outstanding sensitivity toward Cr(VI) with a LOD of 12.2 nM and excellent stability as the signal intensity remain the same after 14 days and decreased by 16.8 % and 30.4 % after 30 days and 5 months, respectively.

2.3.2. Indirect detection

Most heavy metals exist as their ionic states in water, such as Hg^{2+} , Cu^{2+} , Pb^{2+} , Ag^+ etc., which show poor Raman activity. For these ions, molecules with active sites toward specific HMIs have been used to modify traditional SERS substrates. Indirect detection of HMIs can be achieved through the changes in spectral characteristics during the formation of ligand-metal ion complexes.

Liu et al. prepared inositol hexaphosphate protected AgNPs ($\text{IP}_6\text{@AgNPs}$) as a stable SERS substrate, which was then modified with L-cysteine (L-Cys) and Rhodamine 6G (R6G) [69]. The as prepared L-Cys/ $\text{IP}_6\text{@AgNPs}$ aggregated and thus enabled an improvement of SERS signal of R6G. The presence of Cu^{2+} ions could be adsorbed by IP_6 and the concentrated Cu^{2+} ions could further selectively oxidize L-Cys molecules under certain pH value, which dispersed the $\text{IP}_6\text{@AgNPs}$ again and decreased the SERS signal of R6G (Fig. 2B). Due to the specific catalytic reaction of DNAzyme upon binding to Pb^{2+} , DNAzyme is usually used in the SERS system to detect Pb^{2+} . Xu and coworkers developed a sensitive SERS biosensor for highly sensitive detection of Pb^{2+} by using DNAzyme-modified $\text{Fe}_3\text{O}_4\text{@Au@Ag}$ NPs [70]. Cy3-labeled thiolated DNA substrates were used as the SERS probe which were immobilized on the $\text{Fe}_3\text{O}_4\text{@Au@Ag}$ NPs surface, and in the presence of Pb^{2+} , the Cy3-labeled DNA substrates were cleaved by Pb^{2+} -dependent DNAzyme, which caused a remarkable decrease of SERS intensity.

2.4. ECL nanosensor

ECL, also known as electrogenerated chemiluminescence, is a process in which a luminophore generates an excited state by high-energy electron transfer on the electrode surface, and releases light as it returns to the ground state. As the ECL emission mainly involves the energy conversion from electric energy to chemical energy and then to light energy, it avoids light scattering and possesses zero background [71] [72]. Due to the advantages of simple operation, fast response, high sensitivity and zero background, ECL has been widely used in HMIs detection. A typical ECL sensor is generally composed of electrode, luminophore and coreactant, in which electrode modifier, luminophore, coreactant and coreaction accelerator can be nanomaterials.

2.4.1. Nanomaterials as electrode modifiers

An appropriate electrode material is vitally important for developing

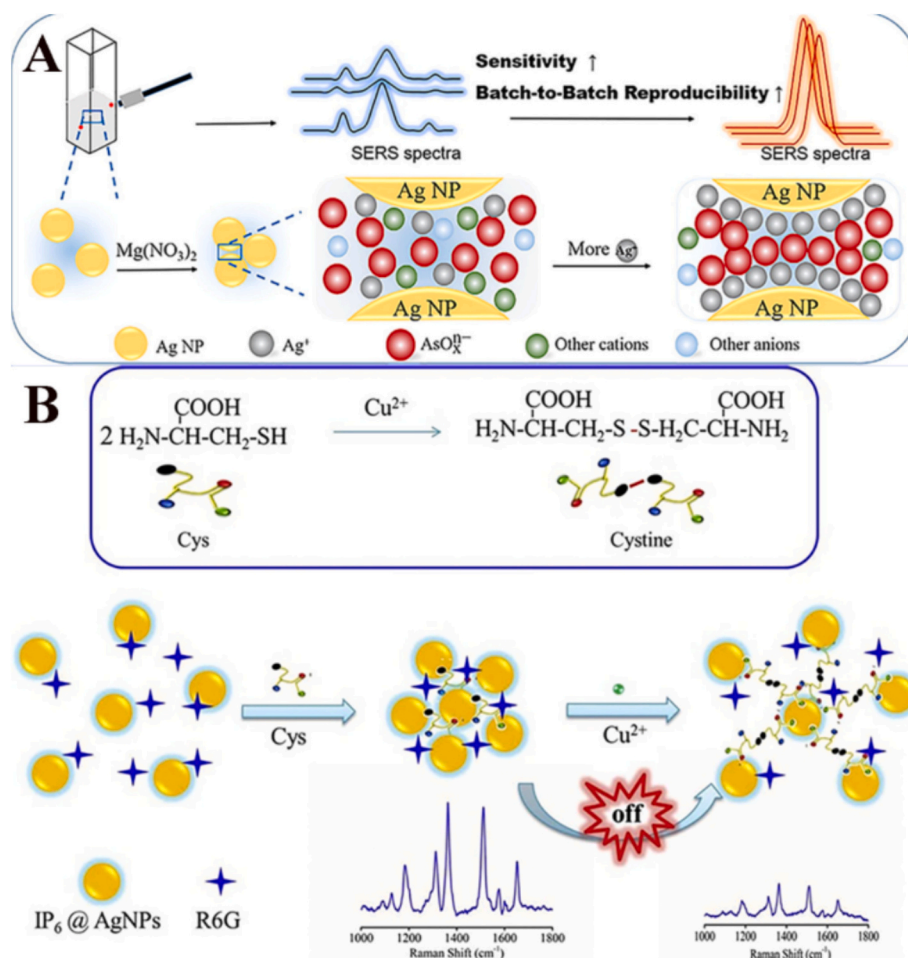


Fig. 2. Schematic illustration of (A) the detection of As(III) and As(V) based on the amplified SERS signal by using AgNPs@Ag⁺ as substrate (Reprinted with permission from [67], Copyright (2022) Elsevier) and (B) the detection of Cu²⁺ based on the decreased SERS signal by the dispersion of IP₆@AgNPs and R6G as the interaction between Cu²⁺ and Cys (Reprinted with permission from [69], Copyright (2019) Elsevier).

ECL-based sensor in HMIs analysis as it is not only the site for emission generation but affects the analytical parameters. Glassy carbon electrode (GCE), gold electrode, indium tin oxide (ITO) electrode, graphite electrode and paper-based electrode are the commonly used electrodes in ECL sensors, which usually need to be modified for obtaining favorable parameters [73]. Up to date, nanomaterials have been reported to modify electrodes.

AuNPs are wonderful electrode modification materials as they can promote the electron transfer and surface area of electrode. AuNPs modified electrodes have been applied in different kinds of ECL sensors and resulted in effective improvement in ECL intensity and sensing sensitivity [74] [75]. In the ECL sensor for detection of Pb²⁺, gold nanodendrites (AuNDs) were assembled onto the ITO electrode. And taking advantage of the large surface area as well as excellent conductivity of AuNDs, the sensor could detect Pb²⁺ quantitatively in a wider range, and performed excellent selectivity [74]. As a result, Au nanomaterials are satisfactory electrode modifiers to obtain highly accurate and efficient ECL sensors for HMIs analysis.

The remarkable electrical, chemical, mechanical and structural properties of carbon-based nanomaterials make them attractive nanomaterials for electrode modification. Carbon nanotubes (CNTs), graphene (G) and graphene oxide (GO) have been reported to modify electrodes of ECL sensors for the detection of various analytes [76] [77] [78] [79] [80]. Due to the superior properties of Au nanomaterials and CNTs, Zhang and Cui developed a simple one-step strategy to synthesize novel hybrid Au nanodots/multi-wall CNTs composite with N-(amino-butyl)-N-(ethylisoluminol) functionalization. The composite was

modified on the ITO electrode to determine Hg²⁺ with a low LOD of 3.2 nM owing to its good electro-transfer property and excellent ECL activity [78].

2.4.2. Nanomaterials as luminophores

ECL luminophores undergo redox processes to produce reactive species that participate in fast electron transfer reactions generating highly energetic species that emit luminescence upon relaxation [72]. The luminescence properties of luminophores are critical determinants that influence their applications in diverse fields. Since the study of Si QDs as luminophores in 2002 [81], novel nanomaterial-based luminophores such as QDs, metal NCs, carbon-based nanomaterials, and MOFs have been widely developed as ECL luminophores for HMIs analysis.

QDs used as ECL luminophores can be divided as compound QDs, core-shell structured QDs and single element QDs [82]. Compound QDs have formed a class of ECL emitters due to their rich electrical and optical properties. However, the most commonly used compound QDs such as CdSe, CdTe, CdS, PbSe etc. are toxic, which seriously restrict their practical applications [83] [84]. Adding protection shells on the surface of compound QDs to form core-shell structured QDs such as CdSe/ZnS, CdS/ZnS, CdTe/CdSe/ZnSe and CdSe/CdS/ZnS can highly improve the biocompatibility [85] [86] [87]. And the core-shell structured QDs have been reported as ECL luminophores for HMIs detection. For example, Wang et al. built a selective ECL sensor for detecting trace amount of Pb²⁺ by using CdSe/ZnS QDs as luminophore coupled with K₂S₂O₈ as coreactant [88]. Similarly, Zhao and coworkers used one-step electro-deposition synthesized CdS/ZnS quantum dots as ECL luminophore and

K₂S₂O₈ as coreactant to detect Cu²⁺ [89].

In addition, carbon-based QDs are low-toxic, water-soluble, biocompatible, environmentally friendly and easy to be functionalized with various agents. And the discovery of ECL behavior of carbon-based QDs is another breakthrough. For example, graphene quantum dots (GQDs) as a zero-dimensional structure, have stronger quantum confinement and boundary effects due to their lateral dimension below 10 nm and exhibit excellent luminescence properties [82]. Tang and coworkers designed an ECL aptasensor by using GQDs as ECL emitter and combining GQDs, AuNPs and single-strand DNA as ECL signal probe for extremely sensitive detection of Hg²⁺ with a detection limit of 2.48 pM [90]. Raju et al. synthesized phosphorous doped carbon quantum dots (PCQDs) as luminophore to study the cathodic ECL properties with H₂O₂ as coreactant. As can be seen in Fig. 3A, more hydroxyl radicals (OH•) were produced with the existence of Cu²⁺, and OH• reacted with PCQDs anion radicals to produce excited state of PCQDs which then emitted the light through reductive oxidation mechanism. The designed ECL platform showed a low detection limit of 0.27 nM toward Cu²⁺ [91].

Metal-organic framework (MOF), with an internal microporous structure, is a composite material self-assembled by organic ligands and metal ions through coordination bonds. However, MOFs are rarely reported as ECL emitters due to their poor optical and electronic conductivity. Interestingly, the optical properties and conductivity of MOFs can be improved by incorporating electroactive moieties into MOFs [93] [94]. For example, Lei et al. prepared electroactive MOFs as ECL emitters. The as prepared electroactive MOFs showed redox properties due to the incorporation of hydroquinone and phenanthroline [93]. In a recent study, Liu and colleagues prepared Ag-MOF as an ECL luminophore for the detection of Hg²⁺ in water samples [94]. Chitosan and AuNPs were used to modify Ag-MOF to improve the ECL stability. A biosensor for the specific detection of Hg²⁺ was established by constructing Hg²⁺ aptamer and steric effect of streptavidin. Upon the introduction of mercury ions, the aptamer was removed and streptavidin could not be immobilized on the electrode. This led to a gradual increase in the intensity of the ECL signal.

2.4.3. Nanomaterials as coreactants

Just like the luminophores, the coreactants also play important roles in ECL sensors. Owing to the strong reducing/oxidizing properties, the intermediates produced by oxidation/reduction of coreactants could react with luminophores to produce ECL signal. Carbon-based QDs have been usually used as coreactants in the self-enhanced ECL sensors in which carbon-based QDs combined with traditional luminophore to form self-enhanced ECL luminophore.

Li et al. investigated the role of carbon nitride quantum dots (CNQDs) as a novel nanomaterial-based coreactant for Ru(bpy)₃³⁺ and then prepared a novel self-enhanced ECL luminophore (Ru-QDs) by

integrating positively-charged Ru(bpy)₃³⁺ and negatively-charged CNQDs via electrostatic interaction [92]. For the as prepared Ru-QDs, the energy loss was significantly reduced and luminous efficiency was enhanced which can be explained by the fact that Ru(bpy)₃³⁺ and CNQDs reacted via an intramolecular electron transfer path because they coexisted in the same nanoparticles (Fig. 3B). The self-enhanced ECL sensor was applied to detect Hg²⁺ based on the different adsorption capacities of T-rich ssDNA and Hg²⁺-triggered dsDNA on the surface of Ru-QDs@SiO₂ nanoparticles. Similar to this, CNQDs and nitrogen doped GQDs (NGQDs) were respectively used as coreactants for Ru(dcbpy)₃³⁺ and Ru(bpy)₃³⁺ for the establishment of self-enhanced ECL sensors to detect Hg²⁺ [95] [96].

2.4.4. Nanomaterials as coreaction accelerators

Coreaction accelerators are species that can significantly enhance the ECL signals through promoting more production of the coreactant intermediates [97]. Nanomaterials as coreaction accelerators have aroused great attention and a vast majority of researches on coreaction accelerators are based on nanomaterials due to their unique chemical and physical properties. Transition metal-based nanomaterials, including single-atom catalysts (SACs), MOFs, noble metal nanomaterials, noble metal-free nanomaterials, and metal oxide nanomaterials, have been used as coreaction accelerators for ECL sensors owing to their excellent catalytic activities, which can accelerate the production of intermediates SO₄^{•−} from S₂O₈^{2−}, reactive oxygen species (ROS) such as superoxide radicals (O₂^{•−}) and OH• from O₂ or H₂O₂, and amine^{•+} from amine [98].

SACs is a type of catalyst in which metal components are dispersed in the form of single atoms on a carrier. With scattered metal active sites and high atomic utilization, O₂ or H₂O₂ can be adsorbed on the active metal center of SACs and efficiently dissociated into ROS during the oxygen reduction reaction (ORR). For example, O₂ or H₂O₂ can be electrochemically reduced by SACs into ROS, and the generated ROS subsequently oxidizes luminol[−] into luminol^{•−} radicals. The generated ROS further reacts with the in situ-formed luminol^{•−} to generate excited-state luminol*, which emits ECL signals upon returning to the ground state [99] [100]. Based on this mechanism, Fe-SACs [100] [101] [102], Co-SACs [99], and Fe/Co-dual SACs [103] have been developed as coreaction accelerators for ECL sensors. With polynuclear sites, MOFs have been reported as coreaction accelerators for ECL sensors. Nd-MOFs exhibited enhanced catalytic activity toward the reduction of S₂O₈^{2−} to SO₄^{•−} and were employed as the coreaction accelerators to significantly promote the ECL emission of the potassium-neutralized perylene/S₂O₈^{2−} system. Coupled with DNAzyme which shows specific catalytic reaction upon binding to Pb²⁺, the ECL sensor was used to detect Pb²⁺ with a LOD of 4.1 fM [104]. Several studies have shown that noble metal nanomaterials can play as coreaction accelerators in ECL sensing

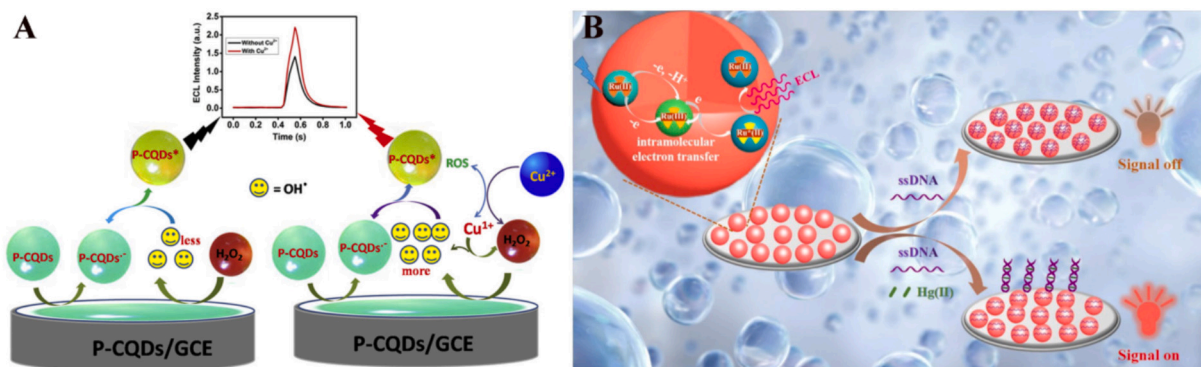


Fig. 3. Schematic illustration of (A) the detection of Cu²⁺ by using PCQDs as ECL luminophore with H₂O₂ as coreactant (Reprinted with permission from [91], Copyright (2020) Elsevier) and (B) the detection of Hg²⁺ by using CNQDs as a novel nanomaterial-based coreactant for Ru(bpy)₃³⁺ (Reprinted with permission from [92], Copyright (2022) Elsevier).

systems [105] [106] [107]. Raju and Kumar modified the surface of GCE with poly(luminol)-gold (PL-Au) nanocomposites to develop a self-enhanced ECL platform without any additional coreactants [107]. The electrochemically deposited AuNPs acted as a coreaction accelerator for the ECL sensor by producing ROS via ORR and the ROS played a crucial role in enhancing the ECL intensity. The presence of AuNPs improved the electron transfer kinetics of poly(luminol) and enhanced the ECL intensity and promoted excellent sensitivity and selectivity toward Hg^{2+} with a low LOD of 0.1 nM. Besides, noble metal-free nanomaterials such as MoS_2 [108], CuS [109], etc. and metal oxide nanomaterials such as TiO_2 [110], Cu_2O [111], V_2O_5 [112], etc. have also been used as coreaction accelerators for ECL sensors.

3. Nanomaterial-based electrochemical sensors

Electrochemical sensors have also attracted great interests in the detection of HMIs owing to the merits of economical cost, fast response time, high sensitivity and easy adaptability for in-situ measurement, and the electrocatalytic activity of the working electrode in a redox system was utilized for HMIs detection [1] [22] [113]. As a result, the working electrode plays a crucial role in the performance of this type of sensor. However, the LOD and the sensitivity of traditional bulk electrodes cannot meet the need for detection of trace HMIs. Fortunately, due to their small size, large accessible surface area, and high catalytic activity, nanomaterials have been used as electrode modifiers or micro/nano-electrodes for the sensitive and selective detection of HMIs.

3.1. Nanomaterials as electrode modifiers

Surface modification with nanomaterials, electrodes show the advantages of higher surface area, improved electron- and mass-transfer rate, and have been extensively employed in the development of novel electrochemical sensors for HMIs detection [22]. In addition, analytical parameters such as LOD, sensitivity, response stability and multi-detection capability are largely improved [1]. Noble metal nanomaterials, noble metal-free nanomaterials, metal oxide nanomaterials and carbon-based nanomaterials have been used as electrode modifiers for the selective and sensitive electrochemical detection of HMIs.

AuNPs show tunable physiochemical properties, large surface areas, higher stability, high electrical conductivity and catalytic activity. Besides, AuNPs can be easily deposited onto the surface of a wide range of electrodes and can decrease the overpotentials of the electroanalytical reactions [1]. Therefore, AuNPs have attracted a great deal of attention in electrochemical sensors for HMIs detection. Researchers reported that AuNPs modified glassy carbon electrodes (GCE) can eliminate the memory effect and interference of coexisting ions, causing lower LOD toward Hg^{2+} , higher sensitivity for As^{3+} and reduced interference effect of Cu^{2+} [114] [115]. In another report, tannic acid (TA)-capped AuNPs modified GCE was prepared for the highly sensitive and selective detection of Hg^{2+} [116]. TA was used as not only a stabilizing agent for AuNPs but a selective capping agent for Hg^{2+} due to the large number of hydroxyl or phenolic hydroxyl groups permitting effective complexation with Hg^{2+} . With the as prepared AuNP@TA-GCE, the electrochemical sensor could selectively detect Hg^{2+} in the presence of competing ions such as Zn^{2+} and Al^{3+} , with a measureable lower limit of 100 fM. At last, AuNP@TA was further modified on disposable screen-printed electrodes (SPEs) to fabricate portable devices which exhibited excellent sensitivity in detecting Hg^{2+} in tap water samples.

Besides AuNPs, other noble metal nanomaterials such as platinum (Pt) or palladium (Pd) nanomaterials have been applied for HMIs detection. For instance, Pt nanoflowers were electrodeposition on bare GCE by using 0.1 M H_2SO_4 solution containing 1.0 mM H_2PtCl_6 for the trace level determination of Pb^{2+} and Cd^{2+} [117]. The produced electrodes have good reproducibility with RSD of 4.65 % for Pb^{2+} and 4.36 % for Cd^{2+} . In another report, PdNPs decorated porous activated carbon materials (Pd@PAC) were prepared via one step thermal reduction

method and modified on GCE for electrochemical detection of trace Cd^{2+} , Pb^{2+} and Cu^{2+} ions. PdNPs provided more active sites and improved conductivity significantly, and with the help of PdNPs, strong current response were generated on the electrode surface [118]. The sensor showed wide linearity in the concentration range of 20–500 nM with individual detection limits of 13.33 nM, 6.60 nM and 11.92 nM for Cd^{2+} , Pb^{2+} and Cu^{2+} , respectively. In addition, the sensor showed high reproducibility with RSD of 1.21 % and long-time stability within 7 days.

Bi-based electrodes are ideal candidate working electrodes as they are environmentally friendly, biocompatible and show a wide cathodic potential window. Inspired by this, BiNPs have been widely used as electrode modifiers for electrochemical sensors. Shi and colleagues constructed sensitive electrochemical sensors for the detection of Pb^{2+} , Cd^{2+} and Zn^{2+} by using novel 3D graphene framework/Bi nanoparticle (GF/BiNP) film-modified GCE as working electrodes [119]. For the preparation of 3D GF/BiNP film, 3D GF was in situ electrochemically reduced and assembled on GCE, and BiNPs were highly dispersed on the framework by a chemical reduction. The as synthesized 3D GF/BiNP film possessed a significantly large active area, fast electron transfer ability, high mass transfer efficiency, and excellent structure stability and binding strength on electrode. Under optimized conditions, the electrochemical sensor displayed ultra low LOD for Pb^{2+} ($0.02 \mu\text{g L}^{-1}$) and Cd^{2+} ($0.05 \mu\text{g L}^{-1}$) and low LOD for Zn^{2+} ($4.0 \mu\text{g L}^{-1}$). In addition, the sensor possessed an excellent stability and the RSDs of 1.76 % for Pb^{2+} and 1.12 % for Cd^{2+} after a long period of 15 weeks were achieved. In another report, Bi modified exfoliated graphite electrode was reported as a simple and cost-effective electrochemical sensor for As^{3+} , Pb^{2+} and Hg^{2+} co-detection. And the as designed electrochemical sensor displayed good stability and reproducibility in HMIs detection in real samples [120]. In a recent study, a novel Bi/ Bi_2O_3 co-doped porous carbon composite (Bi/ Bi_2O_3 @C) was prepared by simple carbonization of Bi-MOFs, which preserved the porous structure of the precursors with strong adsorption, meanwhile, carbon matrix and Bi/ Bi_2O_3 doping were beneficial to electron conduction. Bi/ Bi_2O_3 @C was then modified on the surface of GCE to construct a sensitive electrochemical sensor for Pb^{2+} detection (Fig. 4A). Under optimum conditions, the electrochemical sensor showed a low detection limit of 6.3 nM with good stability (the signal response retains 91 % of the initial value after 7 days), acceptable reproducibility (RSD < 4.6 % ($n = 6$)), and satisfactory selectivity [121].

Owing to their strong adsorption capability and satisfactory electron transfer kinetics, metal oxide nanomaterials have been extensively used in the functionalization of bare electrodes for HMIs detection. For instance, Co_3O_4 nanosheets have been used to modify ITO electrode to fabricate an electrochemical sensor for the detection of Pb^{2+} in real water samples [122]. By simple electrochemical deposition and annealing process, nanostructured Co_3O_4 was modified on the surface of ITO. The as modified electrode was found to be highly selective to Pb^{2+} in the range of 1–100 $\mu\text{g L}^{-1}$, and exhibited a sensitive current response with a LOD of $0.52 \mu\text{g L}^{-1}$. While Co_3O_4 nanomaterials are effective in sensing HMIs, their low electrical behavior is not conducive for high sensitive detection. Therefore, integration of Co_3O_4 nanomaterials with highly conductive substances such as CNTs, GO and reduced GO (rGO) is an effective method to boost their conductivity. In a recent study, You and coworkers developed a novel electrochemical sensor using Co_3O_4 nanocrystals/rGO composite as GCE modifier material to simultaneously detect Cd^{2+} and Pb^{2+} [123]. The incorporation of Co_3O_4 nanocrystals with intense absorption properties resulted in an amplification of the electrochemical current generated on the surface of the sensor by HMIs. Under optimized conditions, the LODs for simultaneous tracking of Cd^{2+} and Pb^{2+} turned out to be 0.062 ppb and 0.034 ppb. In addition, MoS_2 nanosheets have also been used as electrode modifiers to fabricate electrochemical sensors for HMIs detection. Hwang et al. reported an electrochemical sensor by using vertically aligned 2D MoS_2 nanofilm coated Au electrode for in situ Pb^{2+} detection. The vertically aligned 2D MoS_2 sensor showed an excellent linear relationship toward

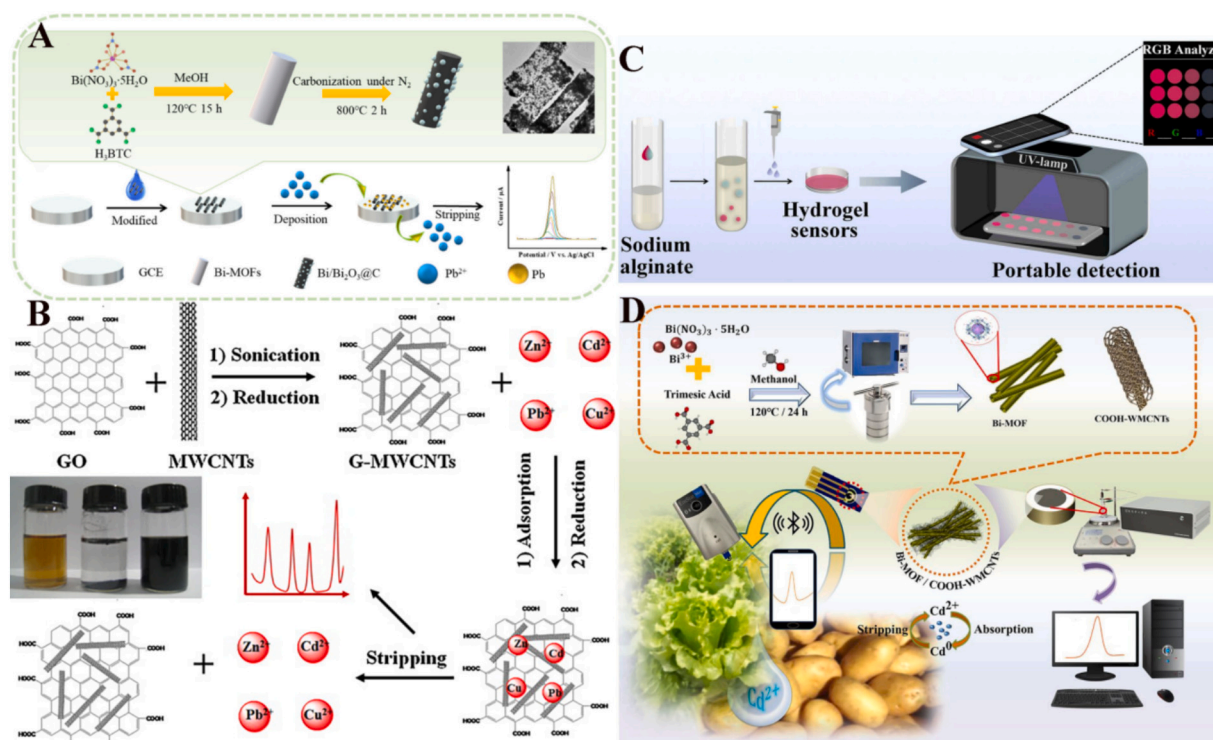


Fig. 4. Schematic illustration of (A) the detection of a Pb^{2+} via a sensitive electrochemical sensor by using $Bi/Bi_2O_3@C$ as the electrode modifier (Reprinted with permission from [121], Copyright (2023) Elsevier), (B) the synthesis of GO-MWCNTs hybrid nanomaterials and their application as novel electrode materials for the electrochemical detection of HMIs (Reprinted with permission from [132], Copyright (2014) Elsevier), (C) the smartphone-assisted $C_3N_4-CdTe_{0.16}S_{0.84}$ QDs hydrogel-based fluorescence sensor for portable on-site analysis of trace Hg^{2+} (Reprinted with permission from [135], Copyright (2024) Elsevier) and (D) the smartphone-based portable sensor coupled with Bi-MOF/COOH-MWCNTs modified SPCEs for electrochemical detection of Cd^{2+} (Reprinted with permission from [136], Copyright (2024) Elsevier).

Pb^{2+} between 0 and 20 ppb with the improved LOD of 0.3 ppb in a tap water environment [124]. Besides, other nanocomposites including MnO_2 [125], ZnO [126], NiO [127], Fe_3O_4 [128], SnO_2 [129], TiO_2 [130], MgO [131] etc. have also been used as electrode modifiers for HMIs detection which will not be elaborated here.

With excellent electronic and optical properties, carbon-based nanomaterials including CNTs, activated carbon, carbon nanofibers, CNT threads, fullerenes, graphene and graphitic substances have emerged as good candidate for HMIs detection [132] [133] [134]. Unmodified CNTs are hydrophobic in nature and insoluble in water and therefore unable to chelate metal ions, which can be overcome by incorporating with other nanomaterials containing carboxylic acid (-COOH), carbonyl (-C=O) or hydroxyl (-OH). Ma and colleagues prepared 3D graphene and MWCNTs (3D G-MWCNTs) nanocomposite as GCE modifier to detect Pb^{2+} and Cd^{2+} [132]. As can be seen in Fig. 4B, the hybrid nanocomposite with excellent conductivity consisted of 1D MWCNTs and 2D graphene nanosheets, which were obtained conveniently by the direct electrochemical reduction of 3D GO-MWCNTs nanocomposites. The as-prepared 3D G-MWCNTs nanocomposites exhibit excellent water-solubility owing to the high hydrophilicity of GO components and were able to track Pb^{2+} and Cd^{2+} ions simultaneously with LOD of 0.1 and $0.2 \mu g L^{-1}$, respectively. In addition, carbon microspheres (CMSs) is a low cost material that can be synthesized easily and possesses good electrical conductivity. Functionalized by amino groups, CMSs were modified on the surface of GCE to selectively and sensitively detect HMIs [133]. The as-designed electrochemical sensor can effectively strip the peaks of Cd^{2+} , Pb^{2+} , Cu^{2+} and Hg^{2+} ions, therefore, these metal ions can be simultaneously or individually detected. The LODs for these metal ions were found to be 1.07, 0.38, 0.25 and $0.37 \mu M$, respectively.

3.2. Nanomaterials as micro/nanoelectrodes

Microelectrode arrays (MEAs) and nanoelectrode arrays (NEAs) have been extensively employed in the electrochemical sensors for HMIs detection to improve the LOD and sensitivity. Compared with conventional macroelectrodes, MEAs and NEAs show higher signal-to-noise ratio, enhanced mass transport, reduced interference effects, and smaller required overpotential [137] [138]. For instance, Gupta et al. prepared a microelectrode that consisted of six highly densified carbon nanotube fibers sectioned in rods (diameter $\sim 69 \mu m$ and length $\sim 40 \mu m$) in a single platform for the determination of trace level of Cu^{2+} , Pb^{2+} and Cd^{2+} in tap water and simulated drinking water [139]. The important contribution of this work is that no additional electrolyte is required for individual and simultaneous electrochemical detection of HMIs. In addition, single Hg nanoelectrodes were fabricated through the controllable electro-deposition of Hg metal on the surface of single gold nanodisk electrodes. Based on the as-prepared nanoelectrodes, a novel nanosensor was established for simultaneous detection of Pb^{2+} and Cu^{2+} with high sensitivity and selectivity, and the limit of detection for Pb^{2+} was $0.021 \mu M$ and for Cu^{2+} was $0.048 \mu M$, respectively [140].

4. Nanomaterial-based electronic sensors

In addition to optical and electrochemical sensors, nanomaterials have also been used in electronic sensors to detect HMIs. FET sensors, a typical electronic sensor, have been used to detect HMIs by utilizing the interaction between the analyte and semiconductor resistor. With high surface-to-volume ratio, 1D and 2D semiconducting nanomaterials such as nanowires, CNTs, graphene, and MoS_2 nanosheets have been extensively used to fabricate FET sensors, which have resulted in high sensitivity. Based on the conductivity change of the semiconducting

nanomaterials in the transistor when the analytes attach to the sensing material surface, HMIs can be detected by the FET sensors.

Si nanowires have aroused attention because that they can be used to fabricate nanoscale devices through a “bottom-up” approach. Si nanowires have been used to construct a FET sensor for the detection of Hg^{2+} and Cd^{2+} with high sensitivity (LOD of 10^{-7} M for Hg^{2+} and 10^{-4} M for Cd^{2+}). The sensing mechanism is based on the accumulation of positive charges on the SiO_2 sheath of Si nanowires which is caused by strong coordination between the heavy metal cations and thiol groups functionalized on the Si nanowires surface as probes [141]. In addition, the high sensitivity to the redox environment of SWCNTs making them candidates for FET sensors. A selective and sensitive FET sensor was constructed with SWCNTs to detect Hg^{2+} based on the conductance change due to the selective redox reaction between SWCNTs and Hg^{2+} . This sensor showed excellent selectivity toward Hg^{2+} over other metal ions along with a LOD of 10 nM and a wide detection range from 10 nM to 1 mM [142]. Owing to its high sensitivity to electronic perturbations caused by foreign molecule adsorption, graphene has been widely used for sensor applications. In the past decades, graphene- and rGO-based FET sensors have been widely used for the detection of HMIs including Hg^{2+} [143], As^{3+} [144], Pb^{2+} [145] [146], Cu^{2+} [147] [148] and Ag^+ [21]. Recently, Du et al. have comprehensively summarized graphene-based FET biosensors for detection of HMIs [149]. They introduced the structures and principles of graphene-based FETs in details, listed examples of graphene-based FETs for detecting HMIs, and analyzed their advantages and disadvantages. Due to its compact structure, aptamer has strict recognition and high affinity with the corresponding target molecules, which makes it more suitable as a probe for graphene- and rGO-based FET sensors. Fang's team has developed a sensor using reduced carboxyl-graphene oxide as FET channel and Pb^{2+} aptamer was directly fixed on the graphene channel with abundant reactive groups. With Pb^{2+} aptamer, the performance of the FET sensor significantly improved with a lower detection limit of 10 pM, a detection range of 10 pM to 100 nM, and a higher interference immunity [145]. MoS_2 , a semiconductor 2D material has been recently reported as a sensing probe in FET sensor for Ag^+ detection [21]. Based on the coordination interactions between Ag^+ and S atoms, the Ag^+ in aqueous solution could spontaneously adsorb on the surface of MoS_2 and affect its electronic carrier transport behavior. The as-designed MoS_2 -based FET sensor show short response time, good linear relationship, low detection limit and high selectivity toward Ag^+ .

5. The combination of nanomaterials with biomaterials for HMIs detection

The interaction of HMIs with biological molecules such as proteins, antibodies, or nucleic acids offers remarkable advantages in terms of selectivity and sensitivity. Based on the specific combination of antibody and antigen, immunosensors have also been used for sensitive and selective detection of HMIs. For example, Kang et al. described a selective and sensitive SPR-based competitive immunoassay via preparing monoclonal antibody against Cd^{2+} to detect Cd^{2+} [150]. Wang et al. developed a chemiluminescent immunoassay strip method for the rapid and sensitive detection of Hg^{2+} , in which an anti- Hg^{2+} monoclonal antibody was synthesized [151]. We can see that antibodies are responsible for sensitive and selective detection of targets. However, the preparation of metal-ion monoclonal antibodies is difficult, expensive and inflating the cost of sensing probes, and polyclonal antibodies have difficulties meeting the specific requirements of metal ions [24].

Up to now, DNAzymes have been studied regarding on their metal binding sites, metal-dependent activity and catalytic mechanisms. One of the most significant practical applications for DNAzymes is HMIs detection due to their high metal ion selectivity [70] [104]. Besides, the catalytic activity of DNAzymes have been utilized for HMIs detection [152]. Promisingly, incorporation of DNAzymes and nanomaterials into sensors for HMIs detection can result in significant improvement in

the performance of sensors in terms of sensitivity, selectivity, multiplexed detection capability and portability. In addition, many researches have been focused on the design of DNA molecules (such as aptamers) functionalized nanostructured sensors to detect HMIs [52] [153]. To date, various nanomaterials, such as graphene or GO [152] [153] [154], QDs [155] [156], AuNPs [157] [158] etc. have been successfully exploited for an effective platform of sensors relying on different signal transduction schemes.

6. Comparison of different sensing approaches

Different sensing platforms based on colorimetry, fluorescence, SERS, ECL, electrochemistry, and electron, were discussed in the previous sections for the detection of HMIs. The key sensing metrics such as detection range, detection limit, response time, reproducibility, stability, and recovery, are outlined in Table S1. From the reported information, all these sensors are sufficiently efficient to detect HMIs.

As can be seen from Table S1, response times are commonly mentioned and reproducibilities and stabilities are rarely studied for colorimetric and fluorescence sensors, and comparable LODs and response times are obtained for these two sensors. In case of SERS sensors, response time, reproducibility, and stability are commonly studied, and lower LODs are obtained comparing with colorimetric and fluorescence sensors. For ECL sensors, stabilities are fundamentally studied and perhaps the lowest LODs are obtained. For instance, the ECL sensor fabricated by Babamiri and coworkers by using dendrimer/ $\text{CdTe}@\text{CdS}$ QDs as probe and AuNPs as quencher exhibits the lowest LOD of 2×10^{-9} nM for Hg^{2+} [159]. Analyzed from the parameters, we can conclude that the ECL sensors demonstrate excellent performances in terms of sensitivity and stability for HMIs detection. For electrochemical sensors, multiplexed and point-of-care analysis can be realized along with superior sensitivity (LODs ranging from 10^{-6} to 10^2 nM), excellent reproducibility, and stability. In the case of electronic sensors, comparable LODs compared with colorimetric and fluorescence sensors are obtained and rapid response times (several seconds) are perhaps the superiority of these sensors.

7. Nanomaterial-based portable sensors

Most of the current conventional sensing techniques for HMIs detection are still dependent on laboratories, which are unsuitable for on-site detection. Therefore, to meet the on-site detection, portable sensors are intensely desired. With the rapid development of nanomaterials and the widespread use of smartphones for everyone, the combination of HMIs sensing services and smartphones will have the potential to offer portable sensing devices for on-site detection.

7.1. Nanomaterial-based portable optical sensors

Red green and blue (RGB)-based portable optical sensors have been widely used for portable on-site measurements in which numerical RGB values corresponding to the sample's color are extracted by the color recognition application from the image that captured by a smartphone's camera and then converted into a measurable optical parameter, such as absorbance or concentration. The optical data can be instantly displayed on the smartphone's screen, providing real-time analysis and results. Up to data, RGB-based strategies have been widely used in colorimetric and fluorescence nanosensors. For portable optical sensors, sensing systems can be performed in liquid solutions [160], on paper strips [14] [161] [162], on hydrogels [135] [163], or on other supports [164]. For example, Luo and coworkers prepared bare nickel selenide (NiSe_2) via a one-step hydrothermal reaction, in which the Se element possesses a strong binding ability with Hg. The as synthesized NiSe_2 exhibited Hg^{2+} -activated oxidase-like activity and oxidized colorless TMB to blue oxTMB. Based on this mechanism, Hg^{2+} concentrations were calculated by the data obtained by UV-Vis spectrophotometer. In order to make the

detection more portable, a colorimetric paper sensor was developed, in which the test papers carrying NiSe_2 were prepared and with the presence of Hg^{2+} and TMB, test papers turned blue which were then converted into RGB values by smartphone. Based on the colorimetric paper sensor, the LOD was calculated to be 8.9 nM and 8.42 nM for different RGB model, indicating that the proposed sensors possess potential application prospects in real water samples [165]. In a recent report, a smartphone-assisted $\text{C}_3\text{N}_4\text{-CdTe}_{0.16}\text{S}_{0.84}$ QDs hydrogel-based fluorescence sensor was constructed for portable on-site analysis of trace Hg^{2+} (Fig. 4C) [135]. The hydrogel carrying the fluorescence probe ($\text{C}_3\text{N}_4\text{-CdTe}_{0.16}\text{S}_{0.84}$ QDs) was prepared by dropping CaCl_2 solution into the mixture of sodium alginate and QDs. After dropping Hg^{2+} solution onto the hydrogel, the fluorescence hydrogel sensors were observed under a UV lamp for visual detection and a color recognizer application was employed to analyze the fluorescence images and quantify the amount of Hg^{2+} .

7.2. Nanomaterial-based portable electrochemical sensors

As known to all, the electrochemical detection is normally performed with a three-electrode system which containing working, reference, and counter electrodes. Screen-printed electrodes (SPEs) are small devices that make the three electrodes assembled onto planar plastic or ceramic substrates, which can be mass-produced owing to screen printing technology. Therefore, SPEs possess the advantages of ease of operation, real-time and on-site sensing ability. In addition, SPEs have become the most suited electrochemical sensors for in situ HMIs detection due to the requirement of small analyte volume and lower power, high sensitivity and shorter response time. The surface modification of SPEs with nanomaterials has greatly improved their sensitivity and selectivity [166] [167] [168]. In a recent study, a portable electrochemical sensor for efficient Zn^{2+} detection was constructed by using $\text{Bi}_2\text{S}_3\text{-MXene}$ nanocomposite modified SPEs [169]. The modified SPEs showed strong affinity for HMIs attributing to Bi_2S_3 and efficient electron transport and excellent electrical conductivity attributing to MXenes. Coupled with a portable electrochemical workstation, the as-modified SPEs were used to on-site detect Zn^{2+} in real tap water and sea water. In another report, a smartphone-based portable sensor coupled with $\text{Bi-MOF/COOH-MWCNTs}$ modified screen-printed carbon electrodes (SPCEs) for rapid electrochemical detection of Cd^{2+} was constructed (Fig. 4D) [136]. Bi-MOF exhibited efficient HMIs detection ability due to the flexible coordination environments of bismuth and the porous advantages of MOFs, while COOH-MWCNTs showed high conductivity. The portable sensor demonstrated good linear range in 0.7–350 ng/mL with a LOD of 0.22 ng/mL for Cd^{2+} . Besides SPEs, a portable magnetic electrochemical sensor was designed for highly efficient detection of Pb^{2+} , in which a magnet attracted magnetic nanomaterials modified GCE was used as the working electrode. The designed portable sensor showed a low detection limit of 0.0043 μM for Pb^{2+} [170].

8. Conclusions and perspectives

In conclusion, various nanomaterials with unique structures and excellent optical, catalytic and electrical properties have been widely used in optical, electrochemical and electronic sensors for HMIs detection. The nanomaterials involved in sensors have greatly enhanced their reproducibility and detection limit for selective determination of HMIs. In this review, we centered on discussing the roles of nanomaterials in optical, electrochemical and electronic sensors for HMIs detection. In addition, the examples of these nanosensors for HMIs detection are provided. Furthermore, to meet the on-site detection, portable sensors are discussed.

Although significant progress has been made in the nanomaterial-based optical, electrochemical and electronic sensors, several challenges remain in the detection of HMIs, which include but are not limited to the following: (i) As the nanomaterials always show high

activity and weak self-stability, the complex real samples (such as seawater, wastewater, food samples, and biological samples) have a severe interference on the sensing signals. To make the nanomaterial-based sensors work efficiently under harsh environmental conditions, future studies should therefore focus on the development of robust sensors, which can also be applicable in portable devices for point-of-care applications. (ii) Due to the sensing mechanisms, non-specific sensing of coexisting substances in complex substrates will occur, which will cause false-positive signal interference. Therefore, sensors with high specificity toward specific HMIs should be designed. (iii) Most nanomaterial-based sensors focus on a single or limited number of HMIs. By designing multiple arrays for different analytes, the applicability of the sensor can be broadened.

CRediT authorship contribution statement

Dong-Mei Liu: Writing – original draft, Project administration, Funding acquisition, Conceptualization. **Chen Dong:** Writing – review & editing, Supervision, Resources. **Hai-Long Jiang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.microc.2025.115511>.

Data availability

No data was used for the research described in the article.

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